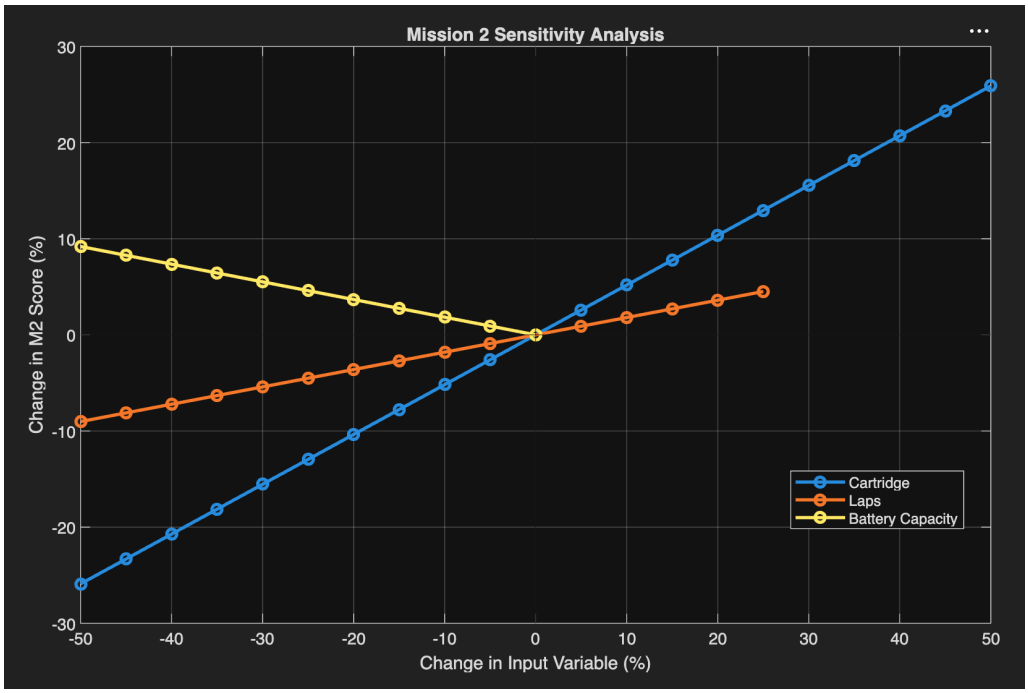
SkyLabs Mission Analysis Task (Write-Up/Analysis): Edward Chang z5769975

Mission 1:
For mission 1, there isn't much to optimize, the team should just aim to pass. Get over 3 laps within 5 minutes to score the 1 point.

Mission 2:



Above: Mission 2 Sensitivity Analysis with constraints applied.
Constraints include:
- Cannot exceed 5 laps.
- Battery cannot exceed 100Wh
- Cartridge count must be less than 27 (generous) or 25-26 realistically is a maximum.

The cartridge count figure was calculated via determining how much mass the plane could carry. It is given that the wing load was $75g\ dm^{-2}$. With the formula

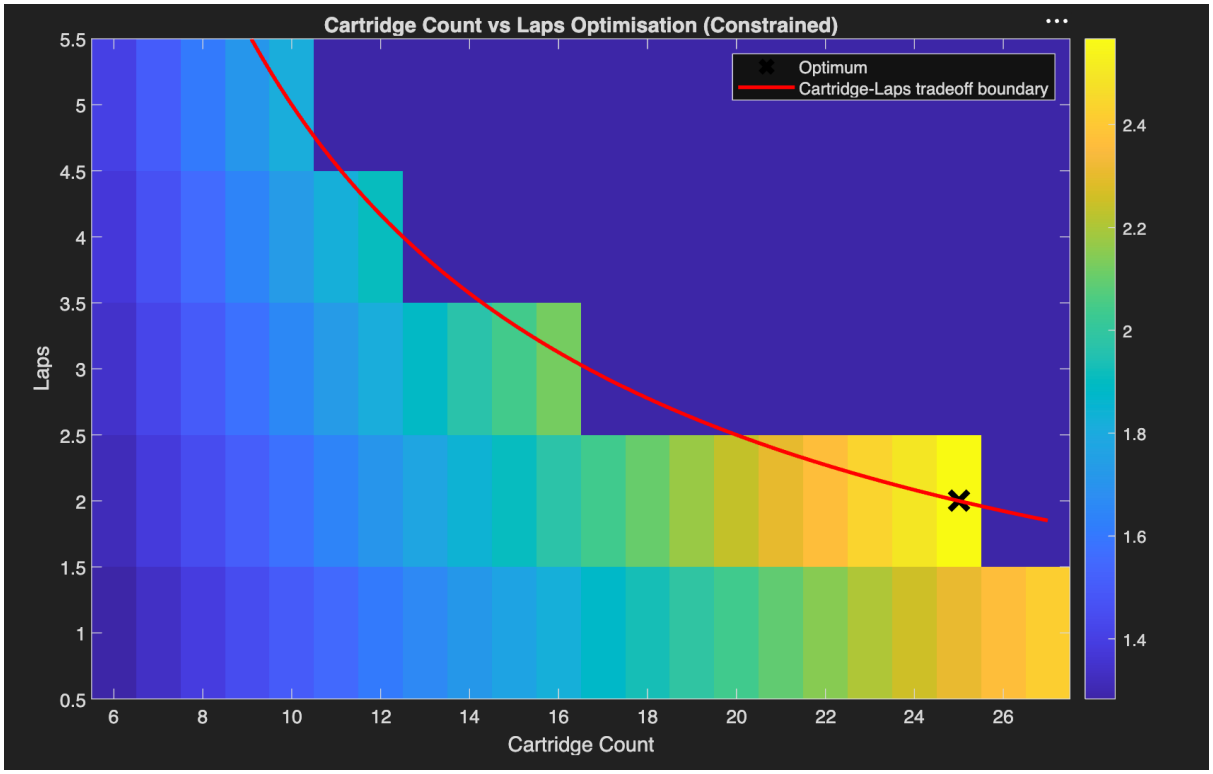
$$Wing\ Load\ = \frac{Total\ Mass}{Total\ Wing\ Area}$$

It could therefore be calculated that the total mass of the aircraft was

$$\text{Wing Load} \times \text{Total Wing Area} = \text{Total Mass}$$
$$0.075kg\ dm^{-2} \times 66\ dm^{-2} = 4.95kg$$

Hence, because the given max weight is 7kg, the payload can be 2.05kg maximum. This value (or 2 kg, to account for supports/bottle/etc) are utilised in the analysis for M2 and M3.

Findings from M2 Sensitivity Analysis:
The above sensitivity analysis therefore shows that increasing cartridge amount is the most beneficial in improving M2 score. Given that battery capacity is already close to the DBF limits at 97.7Wh, there is only a negligible positive extrapolation which cannot be seen.



The above heatmap was made for cartridge count vs laps, utilising a simple constraint curve

$$Laps \propto \frac{1}{Payload\ Mass}$$

which is also used for M3. This is because as laps increases, the payload mass would naturally decrease. However, if we were to derive this relationship from physics, we could utilize forces on an aircraft to make this relationship.

For a plane to fly at a constant altitude (e.g. mostly horizontal lap), lift force must equal to weight, and thrust must equal drag. These can be expressed via the formulas

$$F_{Lift} = \frac{1}{2}\rho V^2 A \cdot C_L = W \ldots (1)$$

And

$$F_{Thrust} = \frac{1}{2}\rho V^2 A \cdot C_D = D \ldots (2)$$

We can therefore see from (1) that $V \propto W^{0.5}$.

By the formula Power = Thrust · Velocity, we can therefore substitute F_{Thrust} into Power.

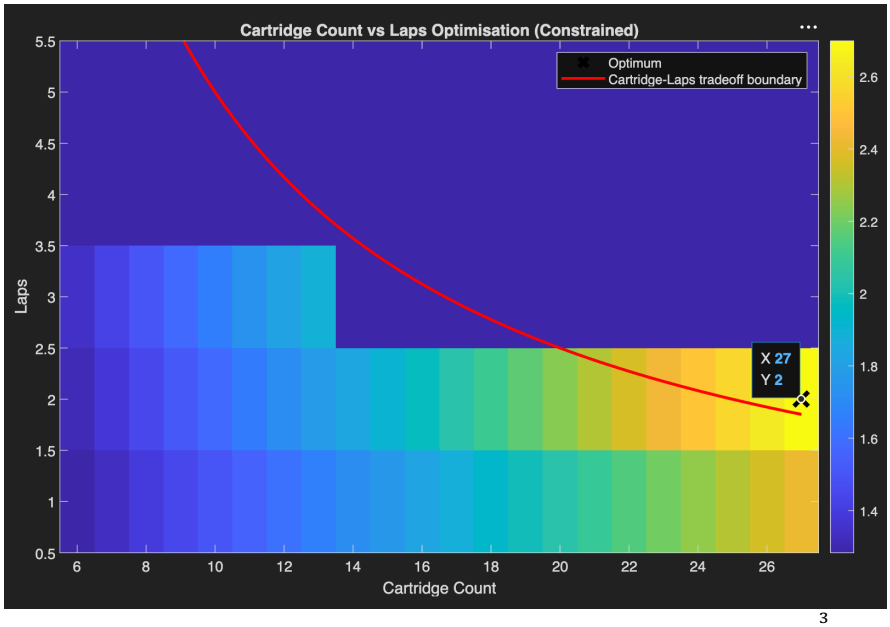
$$P = \left(\frac{1}{2}\rho V^2 A \cdot C_D\right) \cdot V$$

$$P = \frac{1}{2}\rho V^3 A \cdot C_D$$

Because we know that $P \propto V^3$, and that $V \propto W^{0.5}$, we can therefore conclude that $P \propto W^{1.5}$.

Therefore because $L \propto P$, then we can say that $L \propto PayloadMass^{\frac{3}{2}}$.

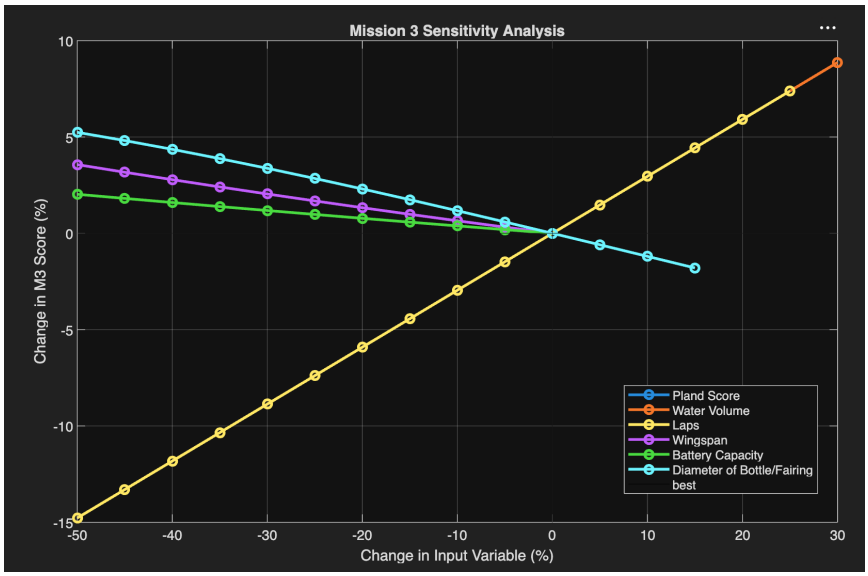
If we apply this curve to the cartridge count vs laps heatmap, we get a similar result.



Above: Cartridge Count vs Laps Heatmap where $L \propto PayloadMass^{\frac{3}{2}}$ *(mass v laps curve)*

However, noting that the additional mass available is 2.05kg for payload INCLUDING the carriers for the cartridge, it is reasonable to state that 27*0.075= 2.025kg of pucks would be rather unachievable (as indicated by the fact that the optimum point is ABOVE the cartridge-laps curve. Therefore, 24-25 cartridges of 0.075kg each would be an optimum amount of cartridges to maximise M2.

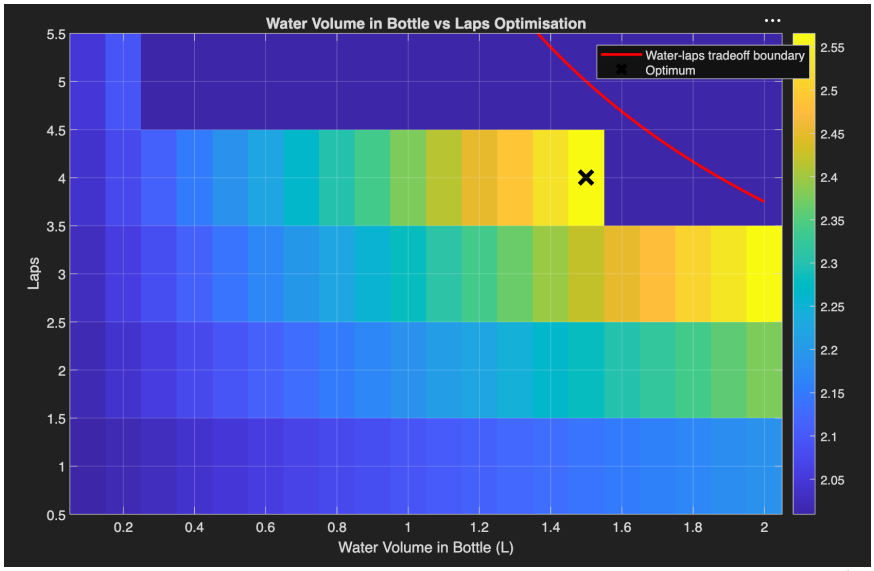
Mission 3:



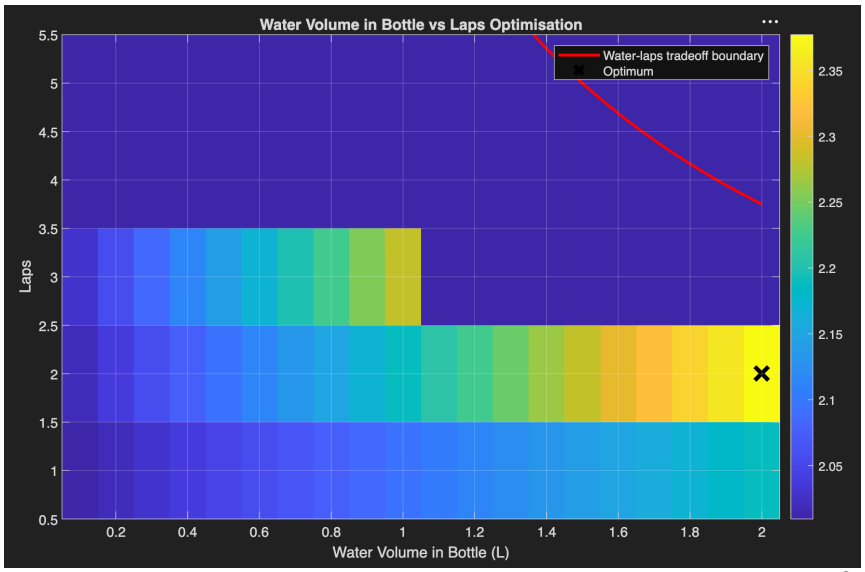
Above: M3 Sensitivity Analysis with Constraints Applied

A notable constraint here was the diameter of the bottle which was capped at 0.1m (based on given information of fuselage size. Wingspan was constrained to under 5 ft (current at 4.7ft)

Findings From M3 Sensitivity Analysis:
We can see that Pland score, Water Volume, and Laps all increase M3 by the same amount, however water volume is the most optimisable hence a focus on water volume is ideal.



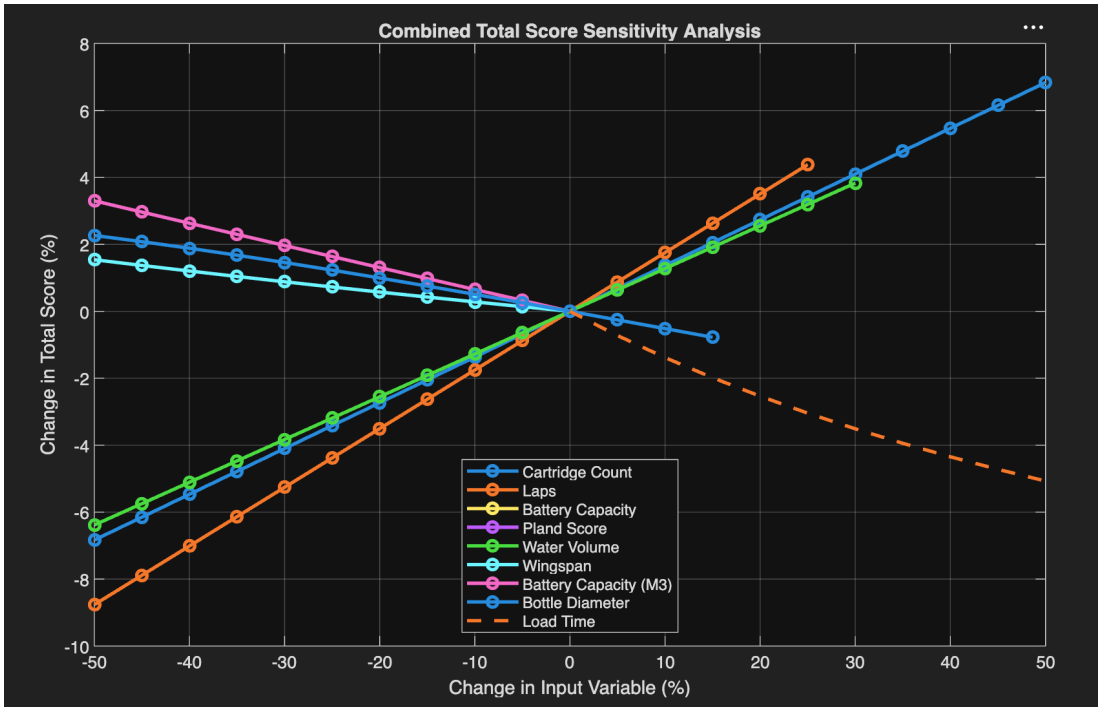
Above: Heat Map of Water Volume to Laps where $L \propto PayloadMass^{-1}$ *(mass v laps curve)*



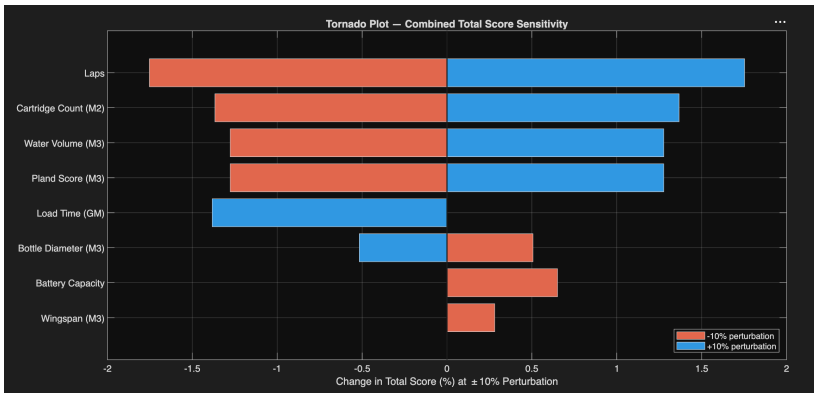
Above: Heat Map of Water Volume to Laps where $L \propto PayloadMass^{-\frac{3}{2}}$ *(mass v laps curve)*

Therefore, taking both models into account, where the second model is more harsh at 2L and 2 Laps (due to increased concavity) and the first model is more lenient at 1.5 Litres and 4 Laps. Plugging in the numbers into M3, model 1 yields a score of 2.566, model 2 yields a score of 2.377, therefore aiming for a model 1 result is more effective in this case, as it also has an optimum point closer to the boundary curve.

TOTAL MISSION + Ground Mission:



Above: Sensitivity Analysis for Total Mission Score



Above: Tornado Plot for Total Score Sensitivity based on Sensitivity Analysis chart values.

Findings from the Total Mission Score Analysis:

The total mission score analysis shows that optimising for cartridge count is highly effective in leading to an improvement in overall mission score, along with the volume of water carried, and laps. This is supported by the tornado chart which is set to change the variables by 10%. However, the volume of water and laps are considerably more limited than cartridge count in the linear chart. Therefore, cartridge count remains to be the highest priority to optimise.

However, it must be noted that the ground mission transition time can significantly impact mission score, as it has a significant decrease of ~-5%, with no constraint. This is also seen in the tornado plot where the effect of load time on the total score slightly surpasses cartridge count. The effect of the ground mission also changes depending on the top team's result, which if significantly lower than our result can significantly decrease mission score (ie. 60 seconds vs 120 seconds is already -0.5).

Hence, the ground mission must also be a significant focus of optimisation, preferably by improving payload accessibility and the ease of loading of the cartridges and bottle.

FINAL FINDINGS:
Therefore, the aircraft on a whole should focus more on being able to lift a greater amount of payload. Across M2 and M3, increases in payload have consistently been the most score-optimising variable. However, as indicated by the tornado plot of total combined score, loading time increase can negatively impact the total score.

Therefore, having a plane with an optimised storage compartment for both storing and inserting/removing payload is essential. A plane with a build similar to the USC's Trojan Horsce from DBF 2026 could potentially perform this duty, given that a cartridges could be stored in the wings. The length of the bottle can also be optimised, which means that diameter does not have to increase whilst the volume of water the aircraft can hold will increase.

$$Total\ Competition\ Score = Total\ Report\ Score * Total\ Mission\ Score + Participation\ Score$$

$$Total\ Report\ Score = 0.15 * Proposal\ Score + 0.85 * Design\ Report\ Score$$

In addition to the mission score, the report score is a multiplier directly on mission score. Therefore, the total report score, and the design report score, must be optimised. A non-negligible amount of effort should be put into the design report in order for the team to score well on a whole in the competition. Participation score is earned at the competition, and provided a prepared team it should not be difficult to get 3/3.